

# YANSHU ZHANG

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## RESEARCH INTERESTS

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I build interactive virtual worlds from images, spanning neural rendering, point-based 3D representations, feed-forward novel view synthesis, 3D reconstruction, editing and animation, and generative models.

## EDUCATION

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**Simon Fraser University**

09/2021 - Expected 04/2027

Ph.D. in Computer Science (CGPA: 4.26/4.3), supervised by Prof. Ke Li

**University of Science and Technology of China**

09/2017 - 07/2021

B.S. in Computer Science

## RESEARCH EXPERIENCE

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**APEX Lab, Simon Fraser University** — Ph.D. Researcher

09/2021 - Present

- Led PAPR (NeurIPS 2023 Spotlight), a point-based neural renderer using ray-to-point proximity attention that deforms without the holes seen in 3D Gaussian Splatting.
- Extended PAPR to geometry/texture editing (PointGT, P-CORE; ECCV 2026), close-up rendering (3DV 2026), 4D interpolation (CVPR 2024 Highlight), and rig-based animation (NeurIPS 2026).
- Built PointWeave, a feed-forward point-based renderer: 31.38 dB PSNR on RealEstate10K, +1.6 dB over 3DGS-based methods, and up to 4× faster than geometry-free models.
- Contributed to generative-model evaluation (a Chamfer-distance metric; NeurIPS 2026) and uncertainty-aware world models for RL (WIMLE; ICLR 2026).

**National University of Singapore** — Research Exchange Programme (non-degree) 01/2021 - 05/2021

## PUBLICATIONS

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PointWeave: Feed-Forward View Synthesis with Explicit Point Interface

**Yanshu Zhang**, Shichong Peng, Chirag Vashist, Ke Li

Under Submission

RigPAPR: Rig-Based Animation of Static Neural Point Clouds from a Single-View Video

Shichong Peng, **Yanshu Zhang**, Ke Li

NeurIPS 2026

The Fault in Our Metrics: Revisiting Generative Model Evaluation with Chamfer Distance

Chirag Vashist, Vaibhav Santhosh, Tristan Engst, **Yanshu Zhang**, Ke Li

NeurIPS 2026 E&D

PointGT: Simultaneous Geometry and Texture Editing for Point-Based Representations

**Yanshu Zhang**, George Shramko, Pratul P. Srinivasan, Ke Li

ECCV 2026

P-CORE: Self-Supervised Surface Consistency for Point-Based Neural Editing

**Yanshu Zhang**, Shichong Peng, Mehran Aghabozorgi, Alireza Moazeni, Ke Li

ECCV 2026

Tackling Misattribution in 3D Intrinsic Decomposition via Proximity Attention Point Rendering

Alireza Moazeni, Shichong Peng, **Yanshu Zhang**, Chirag Vashist, Ke Li

ECCV 2026

PAPR Up-close: Close-up Neural Point Rendering without Holes

**Yanshu Zhang**, Chirag Vashist, Shichong Peng, Ke Li

3DV 2026

WIMLE: Uncertainty-Aware World Models with IMLE

Mehran Aghabozorgi, Alireza Moazeni, **Yanshu Zhang**, Ke Li

ICLR 2026

PROSE: Point Rendering of Sparse-Controlled Edits to Static Scenes

**Yanshu Zhang**, Shichong Peng, Mehran Aghabozorgi, Alireza Moazeni, Ke Li     *ECCV 2024 Workshop*

PAPR in Motion: Seamless Point-level 3D Scene Interpolation

Shichong Peng, **Yanshu Zhang**, Ke Li

*CVPR 2024 (Highlight)*

PAPR: Proximity Attention Point Rendering

**Yanshu Zhang\***, Shichong Peng\*, Alireza Moazeni, Ke Li

*NeurIPS 2023 (Spotlight)*

## PATENTS

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Method and System for Accelerated Operation of Layers Used in a Machine Learning Model and Differentiable Point Rendering Using Proximity Attention

**Yanshu Zhang**, Yuzhen Mao, Shichong Peng, Alireza Moazeni, Ke Li

CA Patent Application: 3292617 (2024)

## TECHNICAL SKILLS

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**Programming:** Python, C++, CUDA (custom kernels)

**Frameworks & Tools:** PyTorch, multi-GPU distributed training (DDP), SLURM clusters

**Research Areas:** Neural rendering, NeRF / 3D Gaussian Splatting, point-based rendering, novel view synthesis, 3D editing and animation, generative models

## AWARDS

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SFU Ph.D. Research Scholarship (2022, 2024)

SFU Graduate Fellowship (2023, 2025)

Silver Award for Outstanding Students of USTC (2019)

Bronze Award for Outstanding Students of USTC (2018)

## TEACHING EXPERIENCE

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Generative Models, Teaching Assistant, SFU (2025)

Machine Learning, Teaching Assistant, SFU (2022)

## ACADEMIC SERVICE

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Served as reviewer for CVPR, ECCV, ICCV, 3DV, SIGGRAPH, SIGGRAPH Asia, ICML, ICLR, NeurIPS